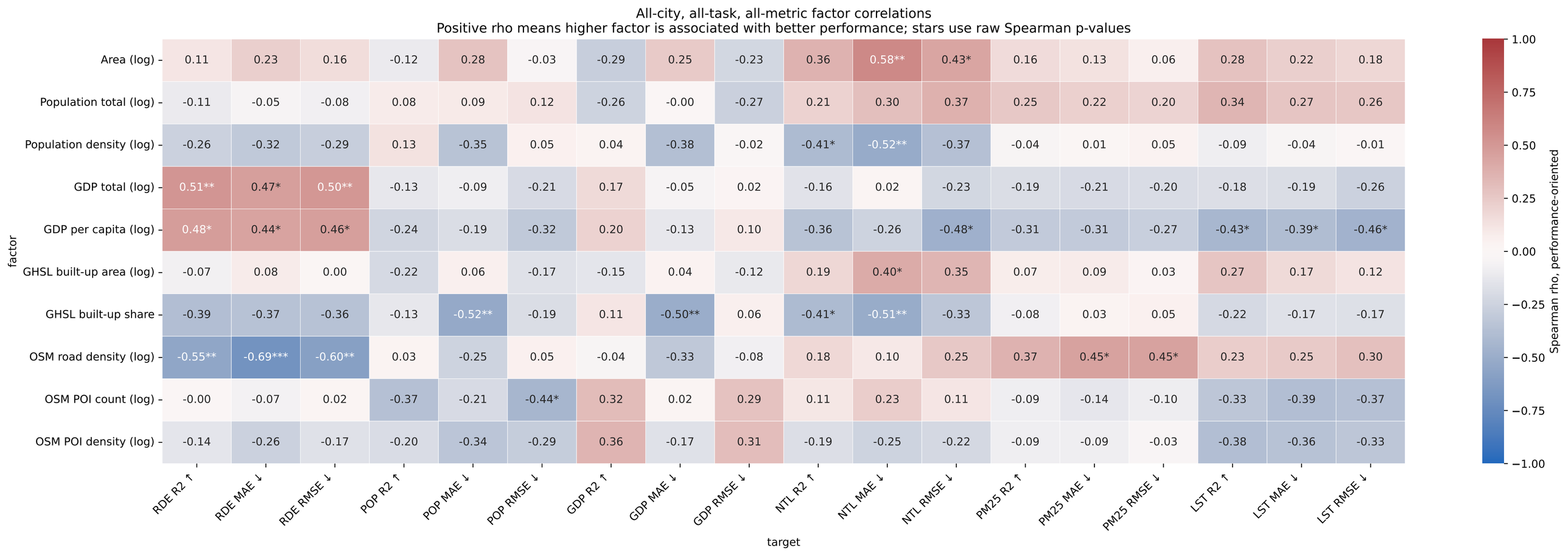
All-city, all-task, all-metric factor correlations
Positive rho means higher factor is associated with better performance; stars use raw Spearman p-values
factor
Area (log)
Population total (log)
Population density (log)
GDP total (log)
GDP per capita (log)
GHSL built-up area (log)
GHSL built-up share
OSM road density (log)
OSM POI count (log)
OSM POI density (log)
0.11 0.23 0.16 -0.12 0.28 -0.03 -0.29 0.25 -0.23 0.36 0.58** 0.43* 0.16 0.13 0.06 0.28 0.22 0.18
-0.11 -0.05 -0.08 0.08 0.09 0.12 -0.26 -0.00 -0.27 0.21 0.30 0.37 0.25 0.22 0.20 0.34 0.27 0.26
-0.26 -0.32 -0.29 0.13 -0.35 0.05 0.04 -0.38 -0.02 -0.41* -0.52** -0.37 -0.04 0.01 0.05 -0.09 -0.04 -0.01
0.51** 0.47* 0.50** -0.13 -0.09 -0.21 0.17 -0.05 0.02 -0.16 0.02 -0.23 -0.19 -0.21 -0.20 -0.18 -0.19 -0.26
0.48* 0.44* 0.46* -0.24 -0.19 -0.32 0.20 -0.13 0.10 -0.36 -0.26 -0.48* -0.31 -0.31 -0.27 -0.43* -0.39* -0.46*
-0.07 0.08 0.00 -0.22 0.06 -0.17 -0.15 0.04 -0.12 0.19 0.40* 0.35 0.07 0.09 0.03 0.27 0.17 0.12
-0.39 -0.37 -0.36 -0.13 -0.52** -0.19 0.11 -0.50** 0.06 -0.41* -0.51** -0.33 -0.08 0.03 0.05 -0.22 -0.17 -0.17
-0.55** -0.69*** -0.60** 0.03 -0.25 0.05 -0.04 -0.33 -0.08 0.18 0.10 0.25 0.37 0.45* 0.45* 0.23 0.25 0.30
-0.00 -0.07 0.02 -0.37 -0.21 -0.44* 0.32 0.02 0.29 0.11 0.23 0.11 -0.09 -0.14 -0.10 -0.33 -0.39 -0.37
-0.14 -0.26 -0.17 -0.20 -0.34 -0.29 0.36 -0.17 0.31 -0.19 -0.25 -0.22 -0.09 -0.09 -0.03 -0.38 -0.36 -0.33
RDE R2 ↑
RDE MAE ↓
RDE RMSE ↓
POP R2 ↑
POP MAE ↓
POP RMSE ↓
GDP R2 ↑
GDP MAE ↓
GDP RMSE ↓
NTL R2 ↑
NTL MAE ↓
NTL RMSE ↓
PM25 R2 ↑
PM25 MAE ↓
PM25 RMSE ↓
LST R2 ↑
LST MAE ↓
LST RMSE ↓
target
Spearman rho, performance-oriented
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00